

Capture the Flag 6: Active or Inactive?

Goal: Make your Student Directory respond to student data by conditionally showing, hiding, and changing UI elements based on each student's state.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding conditional UI.

Create:

`feature/conditional-ui`

Screenshot:

- Terminal showing your current branch

```
→ my_first_flutter_app git:(feature/student-interactions) git checkout -b feat/conditional-ui
Switched to a new branch 'feat/conditional-ui'
→ my_first_flutter_app git:(feat/conditional-ui) |
```

🚩 Flag 1 — Show Active or Inactive

Objective: Change the Student Card based on whether a student is active or inactive.

Add an `active` status to your student data.

Research how Flutter uses **conditional expressions** to display different widgets.

Example states:

- Active
- Inactive

Test

Active students should display an Active indicator.

Inactive students should display an Inactive indicator.

Screenshot:

- Active Student Card
- Inactive Student Card
- Conditional logic in your code



Andrea Reyes

BSIT - 4th Year

Age: 22

Hobby: Digital Art

Student ID: 2022-0045

Email: andrea.reyes@student.edu

Favorite Subject: Multimedia Systems

[Favorite](#) [Activate](#) [Remove](#) [Edit](#)

You have unfavorited this student

This student is not active



Andrea Reyes

BSIT - 4th Year

Age: 22

Hobby: Digital Art

Student ID: 2022-0045

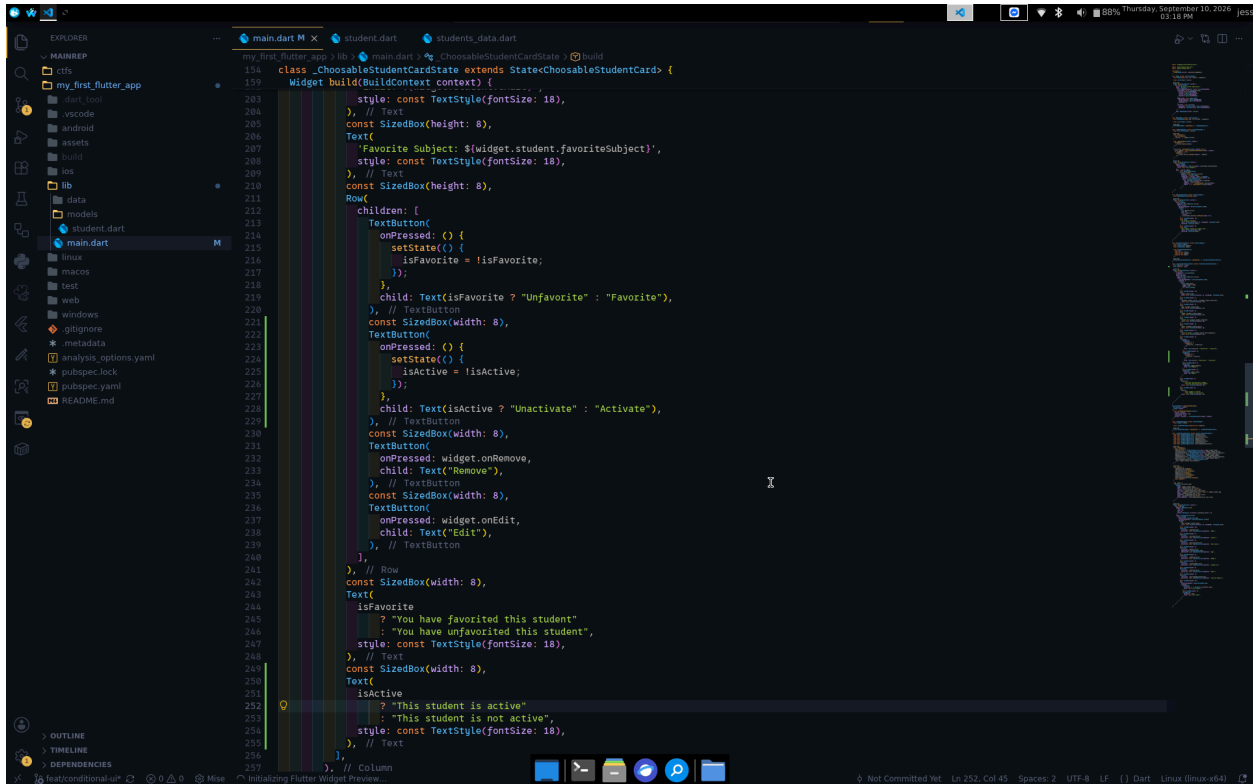
Email: andrea.reyes@student.edu

Favorite Subject: Multimedia Systems

[Favorite](#) [Unactivate](#) [Remove](#) [Edit](#)

You have unfavorited this student

This student is active



Flag 2 — Show or Hide

Objective: Display certain UI elements only when a condition is true.

Add a conditional element to the Student Card.

Examples:

- Favorite badge appears only when favorited
- Warning appears only when inactive
- Delete action appears only if active

Research ways to conditionally include or exclude widgets from the widget tree.

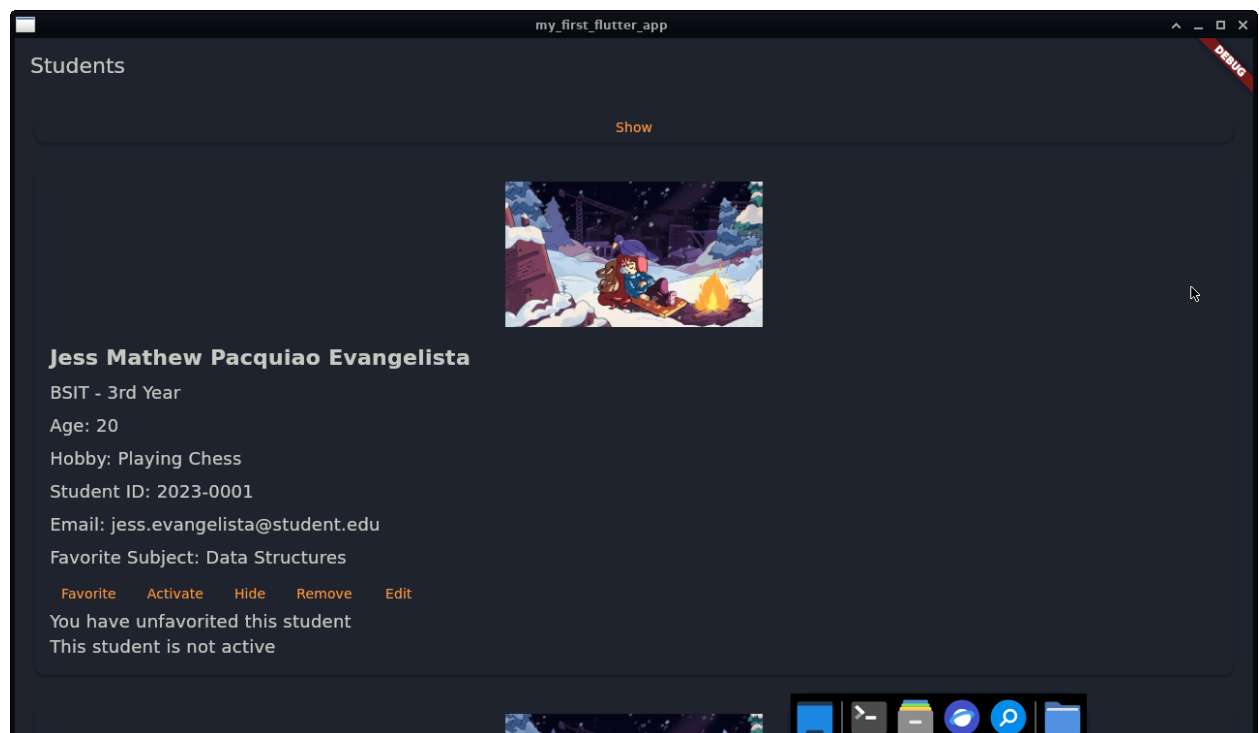
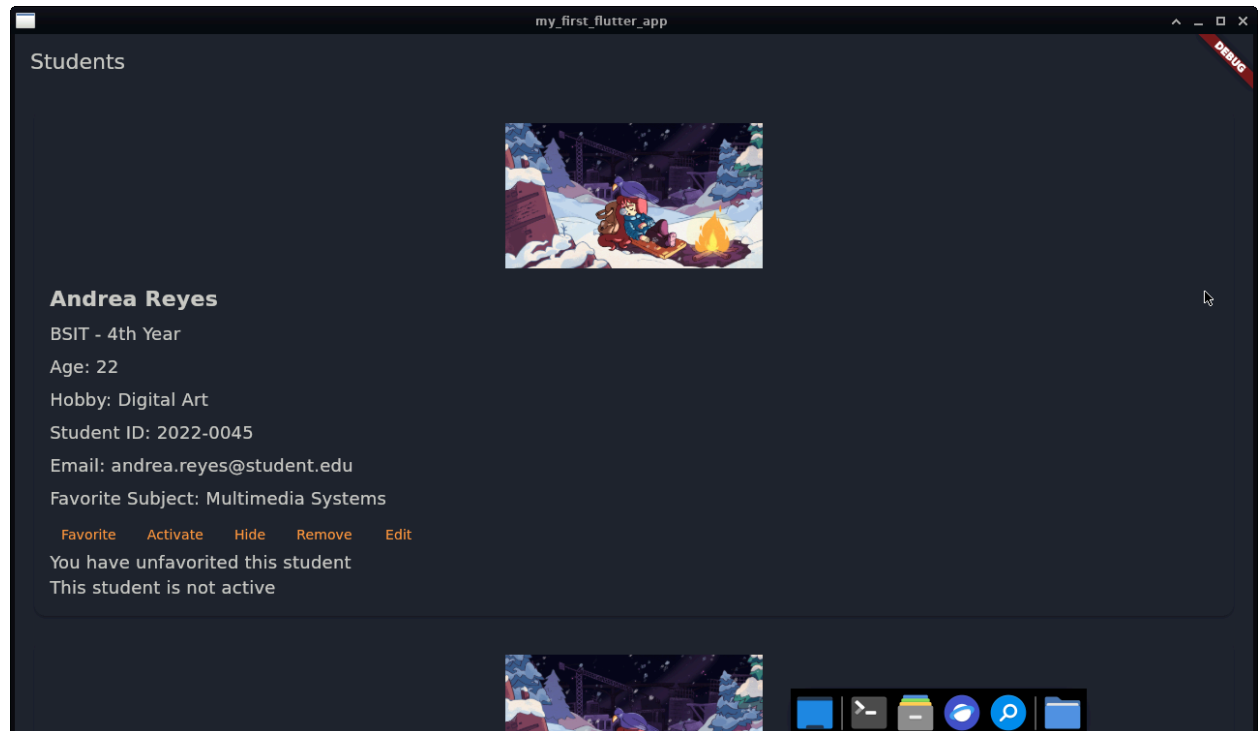
Test

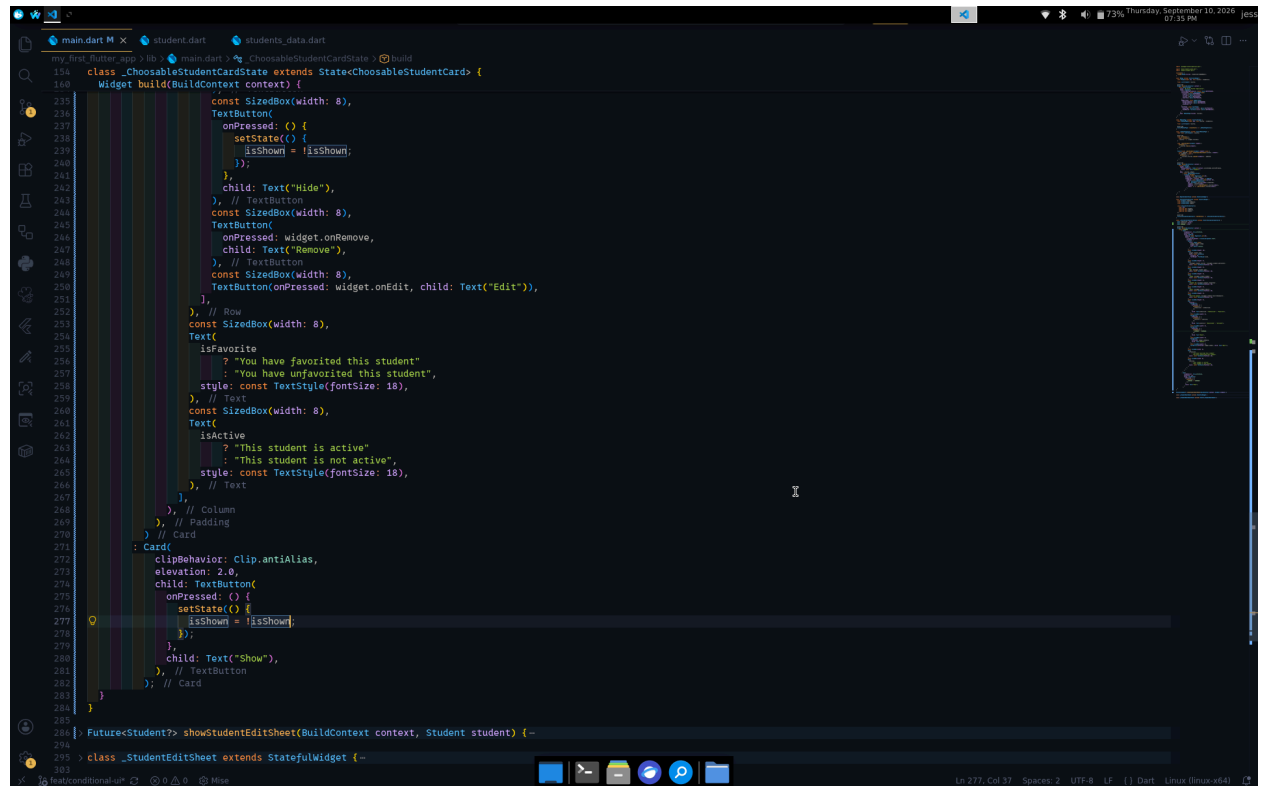
When the condition is true, the element appears.

When the condition is false, the element disappears.

Screenshot:

- UI with the element visible
- UI with the element hidden
- Conditional widget code





```
154 class _ChoosableStudentCardState extends State<ChoosableStudentCard> {
155   Widget build(BuildContext context) {
156     const SizedBox(width: 8),
157     TextButton(
158       onPressed: () {
159         setState(() {
160           isShown = !isShown;
161         });
162       },
163       child: Text("Hide"),
164     ), // TextButton
165     const SizedBox(width: 8),
166     TextButton(
167       onPressed: widget.onRemove,
168       child: Text("Remove"),
169     ), // TextButton
170     const SizedBox(width: 8),
171     TextButton(onPressed: widget.onEdit, child: Text("Edit")),
172   ],
173   // Row
174   const SizedBox(width: 8),
175   Text(
176     isFavorite
177       ? "You have favorited this student"
178       : "You have unfavorited this student",
179     style: const TextStyle(fontSize: 16),
180   ), // Text
181   const SizedBox(width: 8),
182   Text(
183     isActive
184       ? "This student is active"
185       : "This student is not active",
186     style: const TextStyle(fontSize: 16),
187   ), // Text
188   ), // Column
189   // Padding
190   ), // Card
191   Card(
192     clipBehavior: Clip.antiAlias,
193     elevation: 2.0,
194     child: TextButton(
195       onPressed: () {
196         setState(() {
197           isShown = !isShown;
198         });
199       },
200       child: Text("Show"),
201     ), // TextButton
202   ); // Card
203 }
204
205 Future<Student?> showStudentEditSheet(BuildContext context, Student student) {
206   // ...
207 }
208
209 class _StudentEditSheet extends StatefulWidget {
210   // ...
211 }
```

🚩 Flag 3 — The Empty Directory

Objective: Handle situations where there are no students to display.

Modify your Student Directory so it detects when the student list is empty.

Create an **empty state**.

Examples:

- "No students found"
- Empty-state icon
- Button to add a student

Test

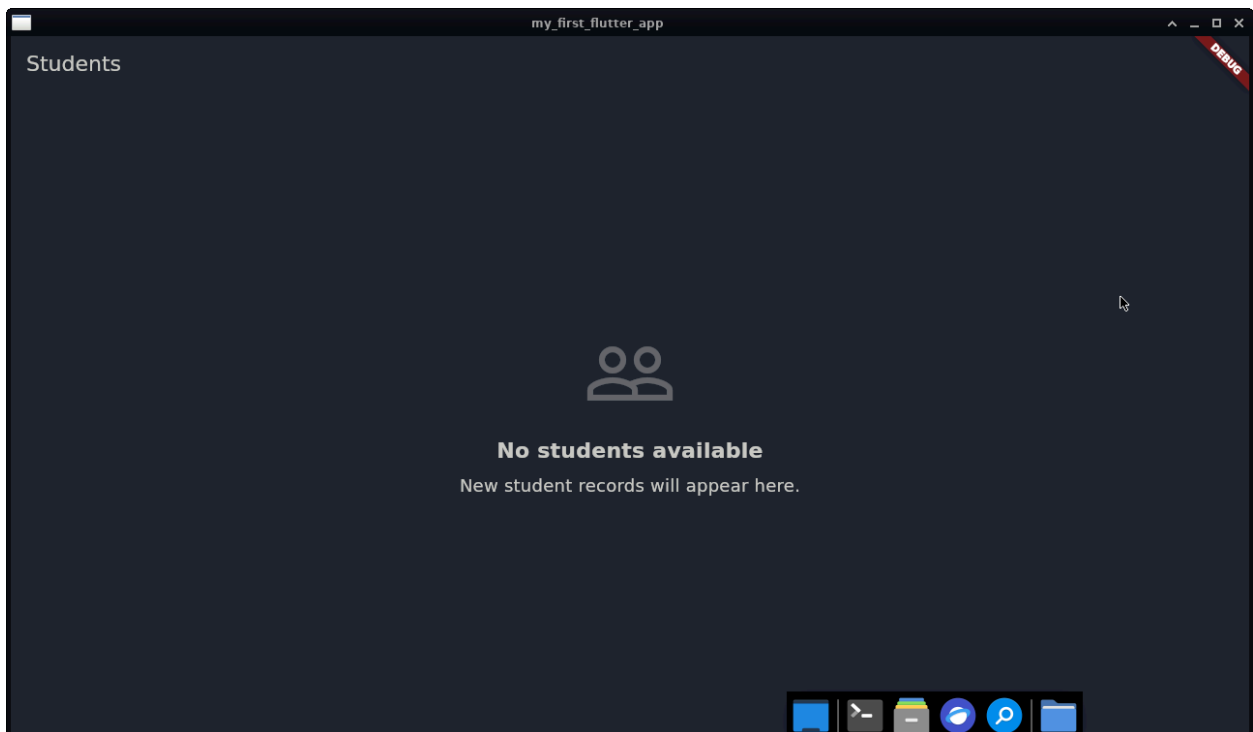
Delete all students in your application (not the code)

Instead of showing a blank screen, your application should display your empty state.

Screenshot:

- Empty student list
- Empty-state UI
- Condition checking for an empty list

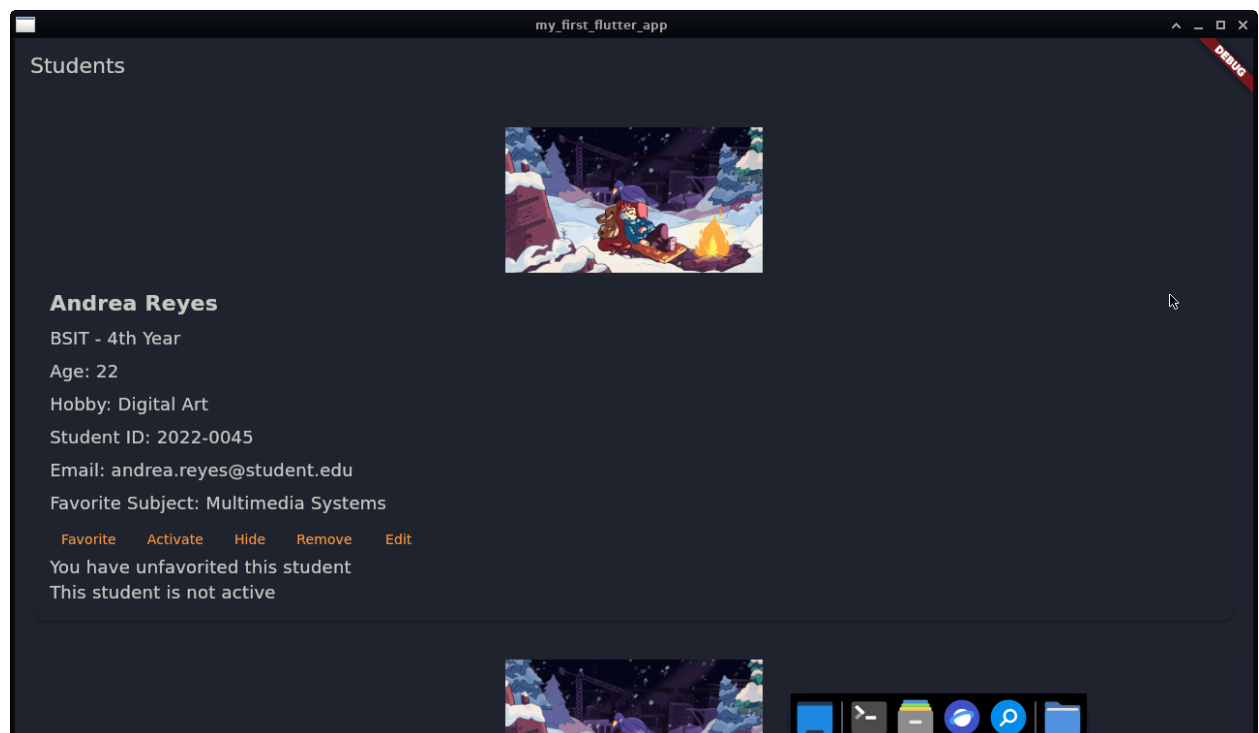
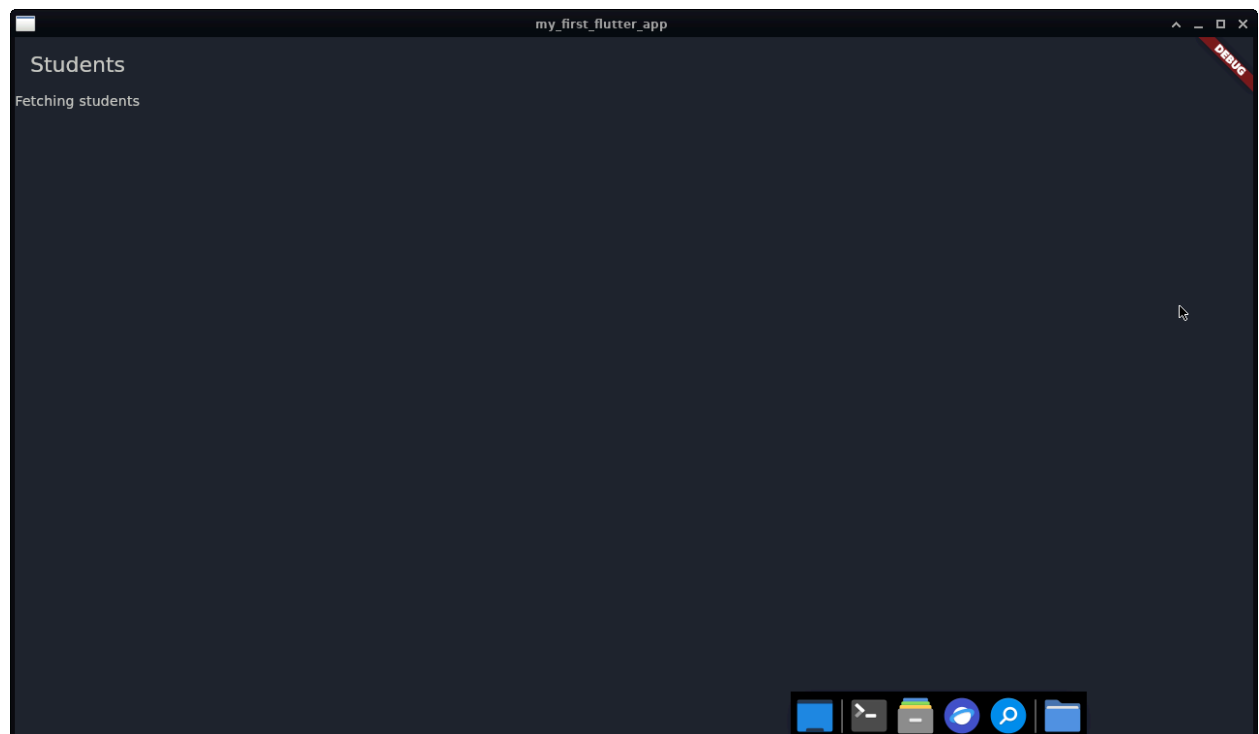
```
main.dart M x student.dart students_data.dart
my_first_flutter_app > lib > main.dart > main
1 import 'package:flutter/material.dart';
2
3 import 'data/students_data.dart';
4 import 'models/student.dart';
5
6
7 void main() {
8   runApp(MyApp(records: []));
9 }
10
11 class MyApp extends StatelessWidget {
12   const MyApp({super.key, this.records = students});
13
14   final List<Student> records;
15
16   @override
17   Widget build(BuildContext context) {
18     return MaterialApp(
19       title: 'My First Flutter Application',
20       theme: ThemeData(
21         scaffoldBackgroundColor: const Color(0xFF1F2A30),
22         colorScheme: const ColorScheme.dark(
23           primary: Color(0xFFFF9940),
24           onPrimary: Color(0xFF1F2A30),
25           surface: Color(0xFF1F2A30),
26           onSurface: Color(0xFFC8CC66),
27         ), // ColorScheme.dark
28         appBarTheme: const AppBarTheme(
29           backgroundColor: Color(0xFF203640),
30           foregroundColor: Color(0xFFC8CC66),
31           elevation: 0,
32         ), // AppBarTheme
33         textTheme: const TextTheme(
34           bodyLarge: TextStyle(color: Color(0xFFC8CC66)),
35           bodyMedium: TextStyle(color: Color(0xFFC8CC66)),
36         ), // TextTheme
37       ), // ThemeData
38       home: MyHomePage(records: records),
39     ); // MaterialApp
40   }
41 }
42
43 class MyHomePage extends StatefulWidget {
44   const MyHomePage({super.key, this.records = students});
45
46   final List<Student> records;
47
48   @override
49   State<MyHomePage> createState() => _MyHomePageState();
50 }
51
52 class _MyHomePageState extends State<MyHomePage> {
53   late final List<Student> _records;
54
55   @override
56   void initState() {
57     super.initState();
58   }
59 }
```

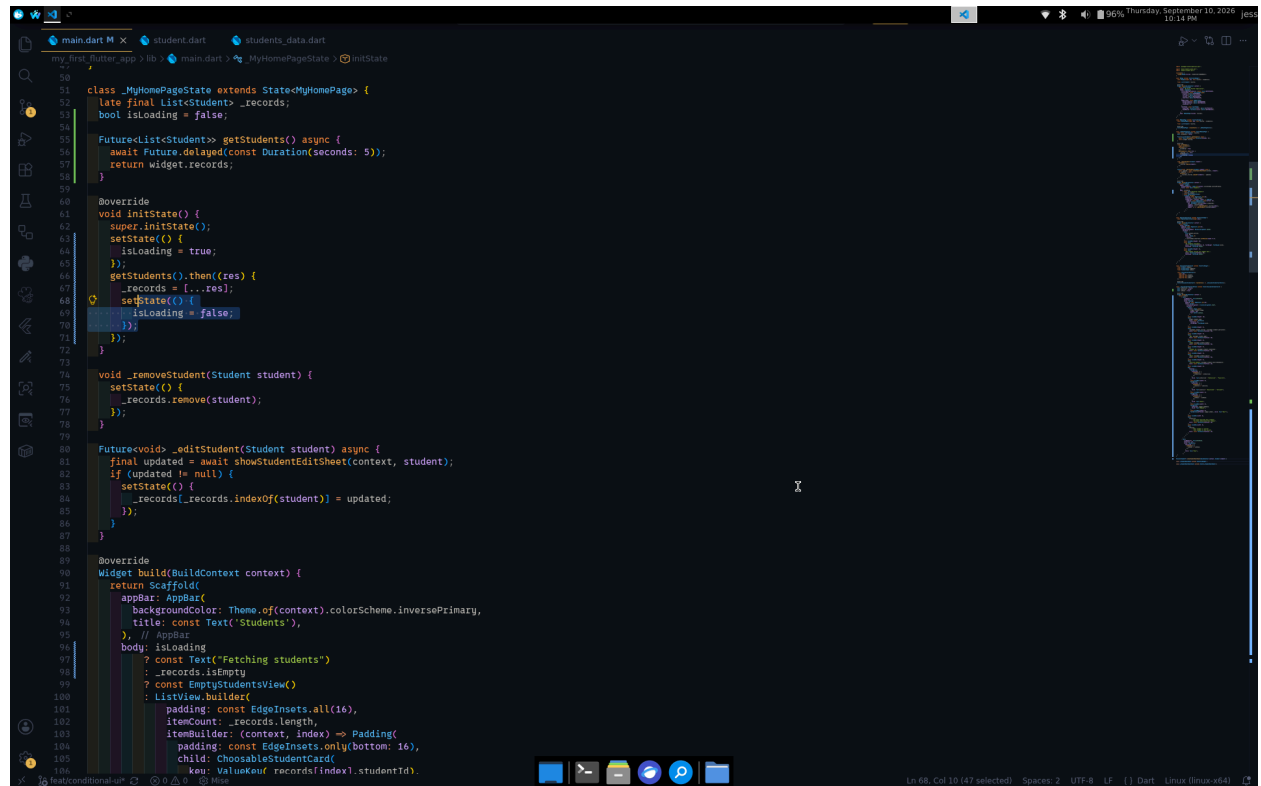


```

51 class _MyHomePageState extends State<MyHomePage> {
52   //
53   //
54   //
55   //
56   Future<void> _editStudent(Student student) async {
57     final updated = await showStudentEditSheet(context, student);
58     if (updated != null) {
59       setState(() {
60         _records[_records.indexOf(student)] = updated;
61       });
62     }
63   }
64   //
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100  //
101  class EmptyStudentsView extends StatelessWidget {
102    const EmptyStudentsView({super.key});
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73
```

- Loaded state
- Conditional loading logic





```
50
51 class _MyHomePageState extends State<MyHomePage> {
52   late final List<Student> _records;
53   bool isLoading = false;
54
55   Future<List<Student>> getStudents() async {
56     await Future.delayed(const Duration(seconds: 5));
57     return widget.records;
58   }
59
60   @override
61   void initState() {
62     super.initState();
63     setState(() {
64       isLoading = true;
65     });
66     getStudents().then((res) {
67       _records = [...res];
68       setState(() {
69         isLoading = false;
70       });
71     });
72   }
73
74   void _removeStudent(Student student) {
75     setState(() {
76       _records.remove(student);
77     });
78   }
79
80   Future<void> _editStudent(Student student) async {
81     final updated = await showStudentEditSheet(context, student);
82     if (updated != null) {
83       setState(() {
84         _records[_records.indexOf(student)] = updated;
85       });
86     }
87   }
88
89   @override
90   Widget build(BuildContext context) {
91     return Scaffold(
92       appBar: AppBar(
93         backgroundColor: Theme.of(context).colorScheme.inversePrimary,
94         title: const Text('Students'),
95       ), // AppBar
96       body: isLoading
97         ? const Text("Fetching students")
98         : _records.isEmpty
99         ? const EmptyStudentsView()
100         : ListView.builder(
101           padding: const EdgeInsets.all(16),
102           itemCount: _records.length,
103           itemBuilder: (context, index) => Padding(
104             padding: const EdgeInsets.only(bottom: 16),
105             child: ChoosableStudentCard(
106               key: ValueKey<records[index].studentId>
```

🚩 Flag 5 — Multiple States

Objective: Make your Student Directory handle multiple UI states correctly.

Your application should now be able to display:

1. Loading
2. Students available
3. No students available

Determine which state should be displayed based on your data.

Test

Verify each state:

Loading

→ Loading UI

Students available

→ Student List

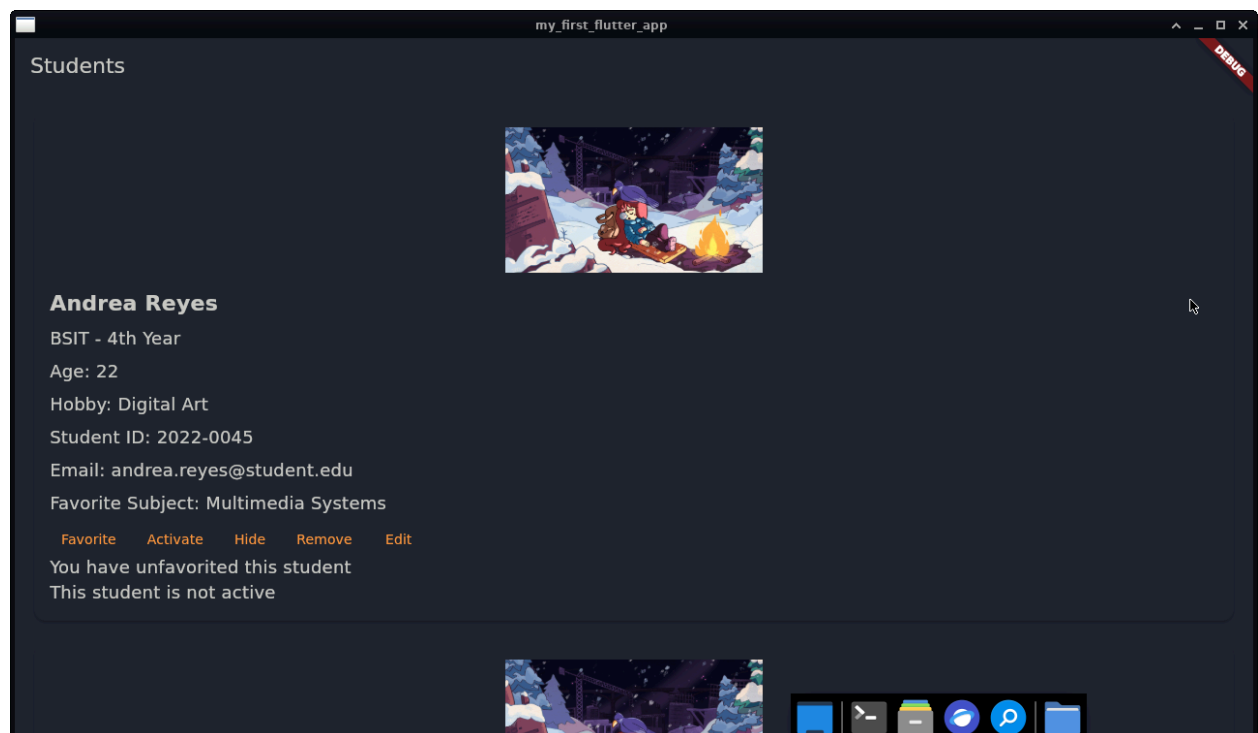
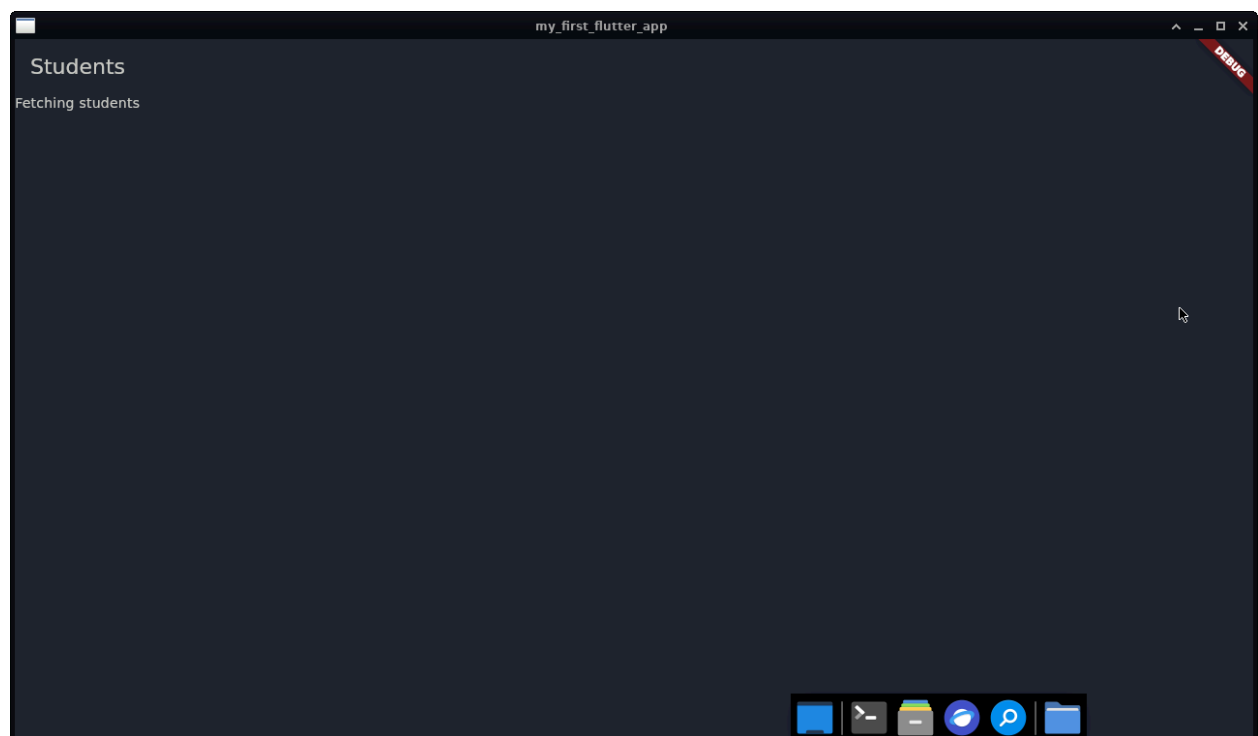
Empty list

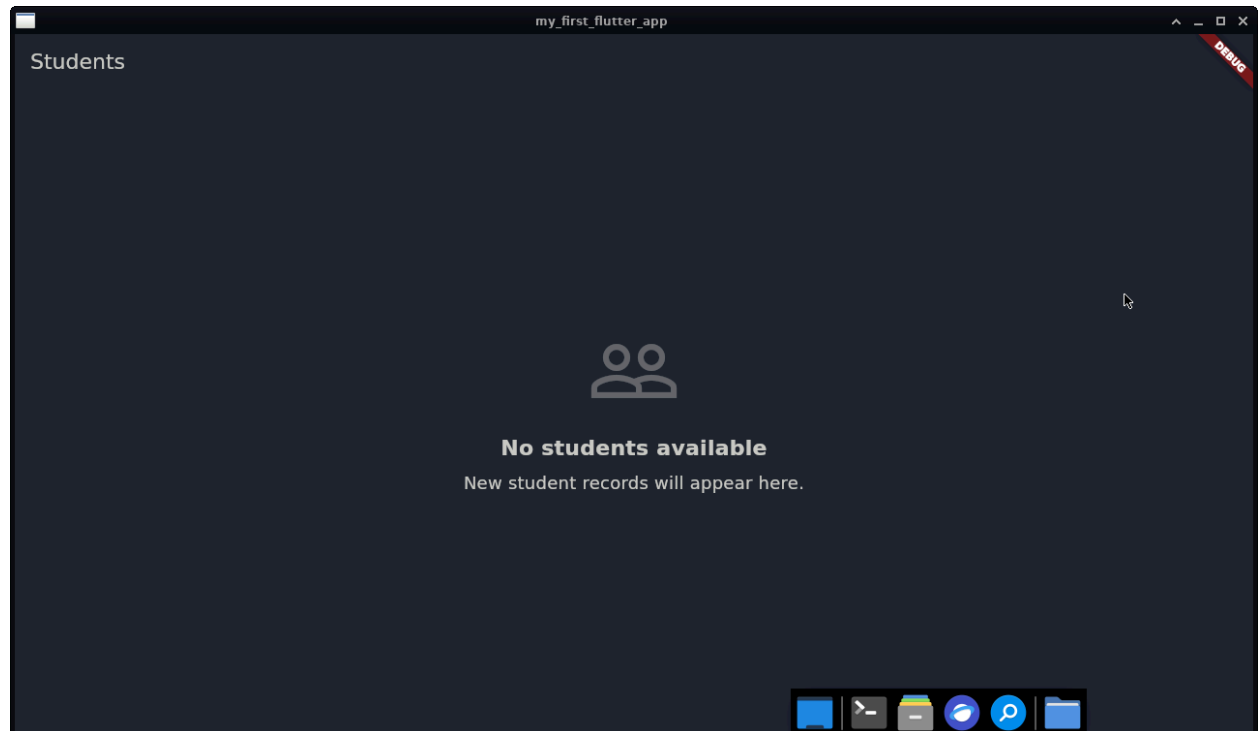
→ Empty State

Only the appropriate UI should appear at a time.

Screenshot:

- Loading state
- Student list state
- Empty state





🚩 Flag 6 — States Work Together

Objective: Make sure conditional UI does not break your existing interactions.

Test the combination of:

- Favorite
- Delete
- Edit
- Active/Inactive
- Show/Hide
- Empty state
- Loading state

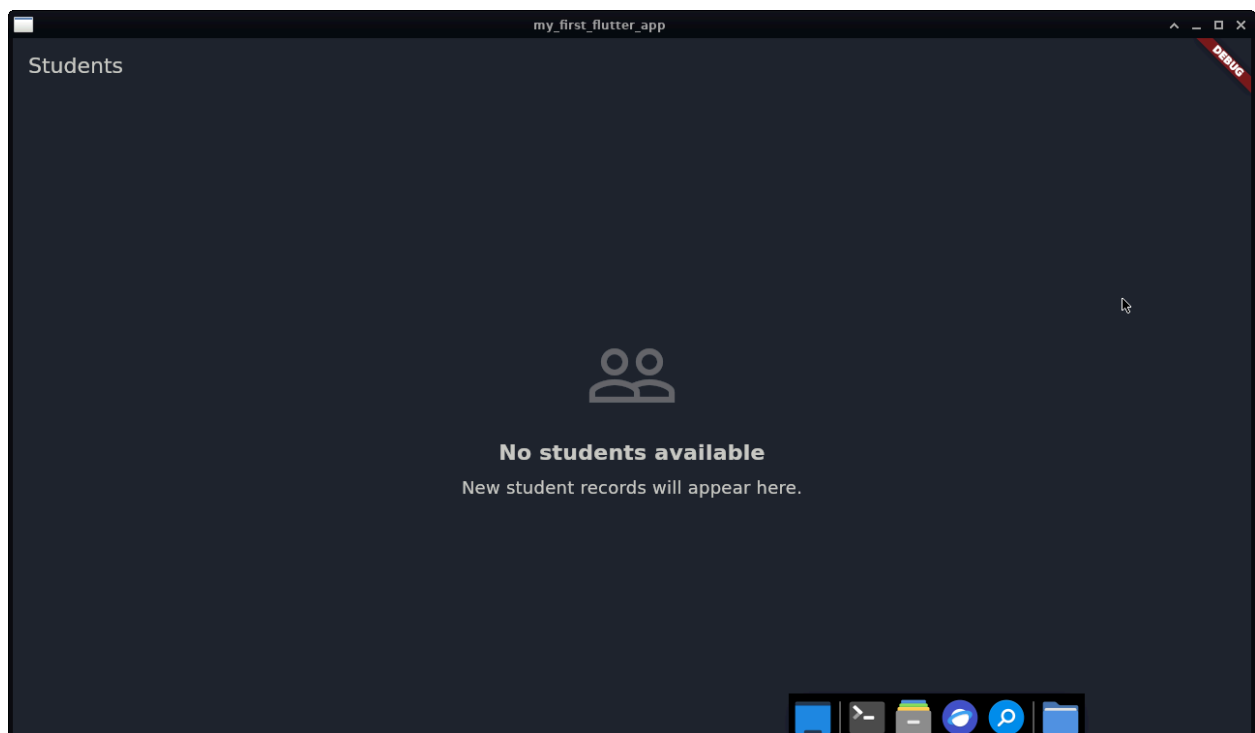
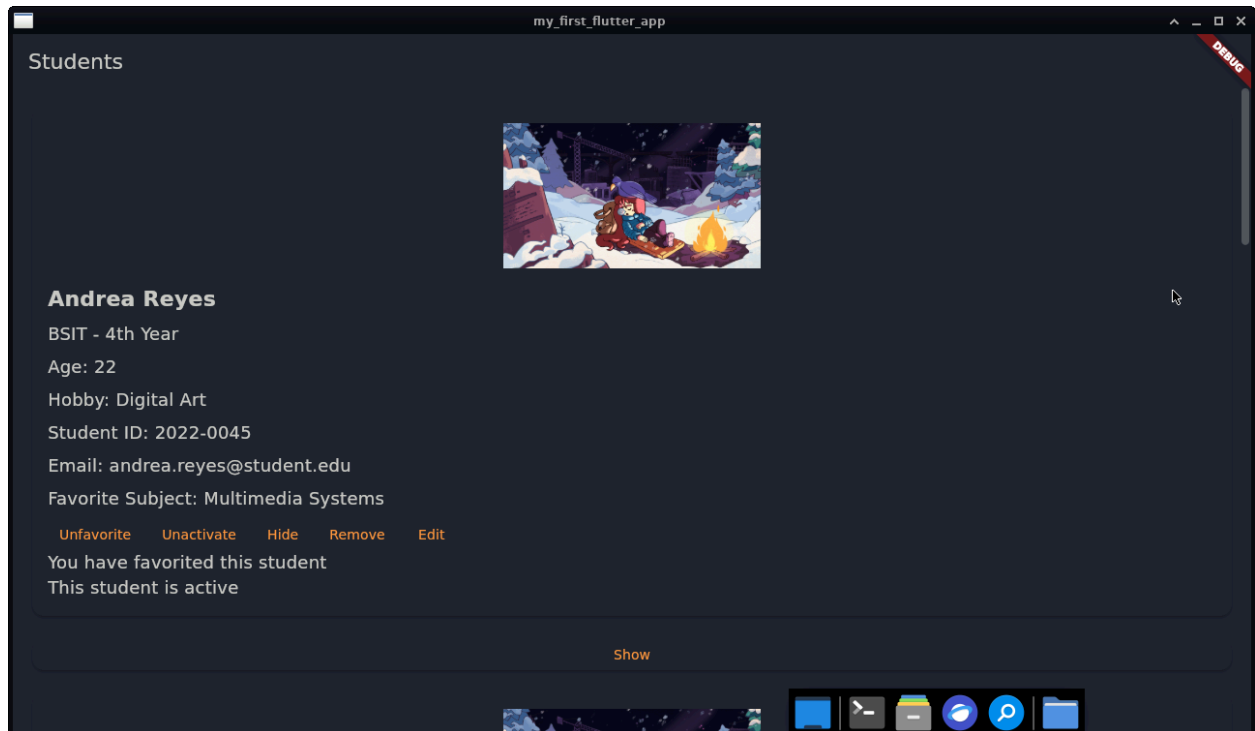
Test Scenario

1. Favorite a student.
2. Change or display their Active/Inactive state.
3. Delete students until the directory becomes empty.
4. Verify the Empty State appears.
5. Reload or restore students.
6. Verify the Student Directory appears again.

The application should continue functioning correctly as its state changes.

Screenshot:

- Multiple conditional states working
- Interactive Student Card
- Empty State
- Application functioning normally



🚩 Flag 7 — Save Your Progress

Objective: Commit and push your completed Conditional UI Student Directory.

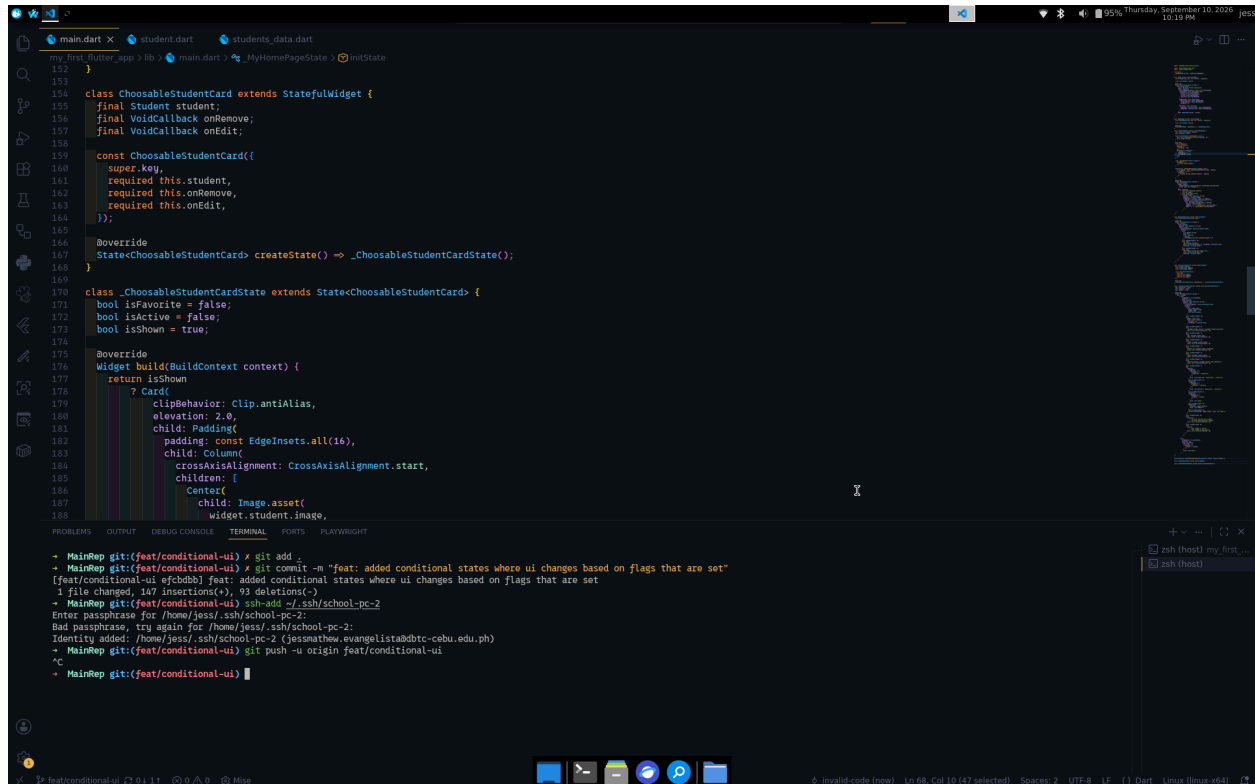
Use a proper Git commit message.

Branch:

feature/conditional-ui

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit



The screenshot shows an IDE with a Dart file named `main.dart` and a terminal window. The Dart file contains the following code:

```
152 }
153
154 class ChoosableStudentCard extends StatefulWidget {
155   final Student student;
156   final VoidCallback onRemove;
157   final VoidCallback onEdit;
158
159   const ChoosableStudentCard({
160     super.key,
161     required this.student,
162     required this.onRemove,
163     required this.onEdit,
164   });
165
166   @override
167   State<ChoosableStudentCard> createState() => _ChoosableStudentCardState();
168 }
169
170 class _ChoosableStudentCardState extends State<ChoosableStudentCard> {
171   bool isFavorite = false;
172   bool isActive = false;
173   bool isShown = true;
174
175   @override
176   Widget build(BuildContext context) {
177     return isShown
178       ? Card(
179         clipBehavior: Clip.antiAlias,
180         elevation: 2.0,
181         child: Padding(
182           padding: const EdgeInsets.all(16),
183           child: Column(
184             crossAxisAlignment: CrossAxisAlignment.start,
185             children: [
186               Center(
187                 child: Image.asset(
188                   widget.student.image,
```

The terminal window shows the following commands and output:

```
+ MainRepo git:(feat/conditional-ui) * git add .
+ MainRepo git:(feat/conditional-ui) * git commit -m "feat: added conditional states where ui changes based on flags that are set"
[feat/conditional-ui 9fcb0bb] feat: added conditional states where ui changes based on flags that are set
1 file changed, 147 insertions(+), 93 deletions(-)
+ MainRepo git:(feat/conditional-ui) ssh-add ~/.ssh/school-pc-2
Enter passphrase for /home/jess/.ssh/school-pc-2:
Bad passphrase, try again for /home/jess/.ssh/school-pc-2:
Identity added: /home/jess/.ssh/school-pc-2 (jessmathew.evangelista@btc-cebu.edu.ph)
+ MainRepo git:(feat/conditional-ui) git push -u origin feat/conditional-ui
^C
+ MainRepo git:(feat/conditional-ui) █
```